

FBISE CLASS 9 MATHEMATICS

CHAPTER 3 - SETS AND RELATIONS

Comprehensive FBISE-Aligned Practice Test

Name: _____	Roll No: _____
Time: 90 Minutes	Total Marks: 75

General Instructions: Attempt all questions. Show complete working where required. Draw neat Venn/arrow diagrams where asked.

SECTION A - OBJECTIVE

Q.1 Choose the correct option. (15 × 1 = 15 marks)

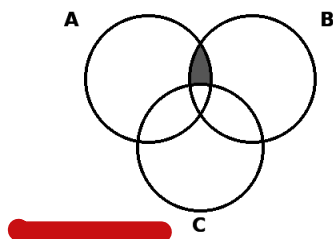
1. Range of $\{(1, 3), (2, 5)\}$ is:

- A. $\{1, 2\}$ B. $\{3, 5\}$
C. $\{1, 3\}$ D. $\{2, 5\}$

2. For $A = \{a, b\}$ and $B = \{1, 2\}$, the relation $R = \{(a, 1), (b, 2)\}$ is a subset of:

- A. $A \times A$ B. $A \times B$
C. $B \times A$ D. $B \times B$

3. In the Venn diagram, the region $A \cap B \cap C'$ represents elements in:



- A. A and B but not in C B. A, B and C
C. A or B but not C D. A only

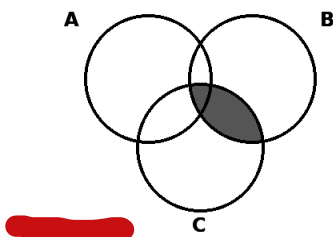
4. What is the domain of the relation $R = \{(1, 2), (2, 3), (3, 4)\}$?

- A. $\{1, 2, 3\}$ B. $\{2, 3, 4\}$
C. $\{1, 3, 4\}$ D. $\{2, 4\}$

5. Given $X = \{2, 4, 6\}$, $Y = \{1, 3, 5\}$ and $R = \{(1, 2), (3, 4), (5, 6)\}$, R is a subset of:

- A. $X \times X$ B. $X \times Y$
C. $Y \times X$ D. $Y \times Y$

6. What statement does the shaded region in the Venn diagram represent?



- A. $A \cap B \cap C$ B. $(A \cap B) \cap C$
C. $B \cap C$ D. $A \cup C$

7. If $A = \{2, 4, 6\}$ and $B = \{0, 1\}$, then the number of elements in $A \times B$ is:

A. 5

B. 6

C. 8

D. 9

SECTION A - OBJECTIVE (Continued)

Q.1 Choose the correct option.

8. If $n(A) = p$, then $n(A \times A) =$

- A. p
- B. $2p$
- C. $2p^2$
- D. p^2

9. If $R = \{(a, b) \mid a, b \in \mathbb{N} \text{ and } a \times b = 12\}$, then the tabular form of R is:

- A. $\{(1,12), (2,6), (3,4), (4,3), (6,2), (12,1)\}$
- B. $\{(1,12), (2,6), (3,4)\}$
- C. $\{(12,1), (6,2), (4,3)\}$
- D. $\{(1,12), (4,3), (2,6)\}$

10. If $R = \{(a, b) \mid a, b \in \mathbb{Z} \text{ and } a + b = 0\}$, then:

- A. $\text{Dom } R \subset \text{Range } R$
- B. $\text{Range } R \subset \text{Dom } R$
- C. $\text{Dom } R = \text{Range } R$
- D. None of these

11. The number of different ways commonly used to describe a set is:

- A. 1
- B. 2
- C. 3
- D. 4

12. The set $\{x \mid x \in \mathbb{W} \text{ and } x \leq 101\}$ is:

- A. Infinite set
- B. Subset
- C. Null set
- D. Finite set

13. The set-builder form of $A - B$ is:

- A. $\{x \mid x \in A\}$
- B. $\{x \mid x \in A \text{ and } x \notin B\}$
- C. $\{x \mid x \in A \text{ and } x \in B\}$
- D. $\{x \mid x \in B\}$

14. If $A \cup B = A$ and $A \cap B = B$, then:

- A. $A \subseteq B$
- B. $A \not\subseteq B$
- C. $A \supseteq B$
- D. $A \neq B$

15. If $A \subset B$, then $A - B =$

- A. A
- B. B
- C. $\emptyset$
- D. $B - A$

SECTION B - SHORT QUESTIONS

Q.2 Answer all of the following short questions. (15 × 4 = 60 marks)

- (i) If $A = \{a, b, c, d\}$, $B = \{a, c, e\}$ and $U = \{a, b, c, d, e, f\}$, show using a Venn diagram that $(A \cup B)' = A' \cap B'$.
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- (ii) Let $A = \{1, 4, 8\}$ and $B = \{1, 6\}$. Find the Cartesian product $A \times B$. Also find $R = \{(x, y) \mid x \in A, y \in B \text{ and } x \geq y\}$.
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- (iii) Using a Venn diagram, prove that $(P \cup Q) \cup R = P \cup (Q \cup R)$, where $P = \{0, 1, 2, 3\}$, $Q = \{2, 3, 4, 5, 6\}$ and $R = \{5, 6, 7, 8, 9\}$.
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- (iv) If $A = \{1, 3\}$ and $B = \{2, 4\}$:
(a) Write the Cartesian product $A \times B$.
(b) Find $R_1 = \{(x, y) \mid x \in A, y \in B \text{ and } x > y\}$.
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- (v) If $X = \{1, 3, 9\}$, $Y = \{3, 5, 7\}$ and $Z = \{3, 5, 7, 9, 11\}$, then using a Venn diagram prove that $X \cup (Y \cap Z) = (X \cup Y) \cap (X \cup Z)$.
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- (vi) For $A = \{1, 2, 3\}$ and $B = \{3, 4\}$:
(a) List all ordered pairs of $A \times B$.
(b) List all ordered pairs of $R = \{(x, y) \mid x \in A, y \in B \text{ and } x < y\}$.
(c) Find the domain and range of R .
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SECTION B - SHORT QUESTIONS (Continued)

Q.2 Answer all of the following short questions.

- (vii) Verify using a Venn diagram that $A \cap (B \cap C) = (A \cap B) \cap C$ for $A = \{2, 3, 5, 7\}$, $B = \{1, 2, 3, 5\}$ and $C = \{3, 4, 6\}$.
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- (viii) For $X = \{1, 9\}$ and $Y = \{2, 4, 6\}$:
- (a) Find $X \times Y$.
- (b) List all ordered pairs of $R = \{(x, y) \mid x \in X, y \in Y \text{ and } x > y\}$.
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- (ix) Let $A = \{1, 2, 3\}$ and $B = \{4, 5\}$. Find $A \times B$ and write the elements of its domain and range.
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- (x) Use a Venn diagram to verify that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$, where $A = \{1, 2\}$, $B = \{2, 3\}$ and $C = \{1, 3\}$.
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- (xi) Verify using a Venn diagram that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ for $A = \{1, 2, 3, 4, 5\}$, $B = \{2, 5, 7, 9\}$ and $C = \{3, 5, 9, 11\}$.
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SECTION B - SHORT QUESTIONS (Continued)

Q.2 Answer all of the following short questions.

- (xii) For $A = \{1, 3, 5\}$ and $B = \{4, 5\}$:
- (a) Find $A \times B$.
 - (b) List all ordered pairs of $R = \{(x, y) \mid x \in A, y \in B \text{ and } x + y \geq 7\}$.
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- (xiii) Let $A = \{1, 2, 3\}$, $B = \{0, 1, 3\}$ and $R = \{(a, b) \mid a \in A, b \in B \text{ and } a > b\}$. Find R and represent it by an arrow diagram.
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- (xiv) Given $A = \{0, 2, 3\}$, $B = \{0, 2, 4, 6, 9, 16\}$ and $R = \{(x, y) \mid x \in A, y \in B \text{ and } x^2 = y\}$, verify that $\text{Dom}(R^{-1}) = \text{Range}(R)$ and $\text{Range}(R^{-1}) = \text{Dom}(R)$.
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- (xv) Let $A = \{0, 1, 3\}$ and $B = \{1, 2, 3, 5, 7\}$. Write $R = \{(x, y) \mid x \in A, y \in B \text{ and } y = 2x + 1\}$ in tabular form. Also find R^{-1} .
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End of Test - Recheck your MCQ choices, set notation, ordered pairs, domains/ranges, and diagrams before submission.